



Yakeen Plus (2025)

Practice Test - 04

DURATION : 200 Minutes

DATE : 09/10/2024

M. MARKS : 720

Topics Covered

Physics :	Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids
Chemistry :	Electrochemistry, Chemical Kinetics, Structure of Atom
Biology :	(Botany): Biological Classification (Zoology): Excretory Products & their Elimination

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

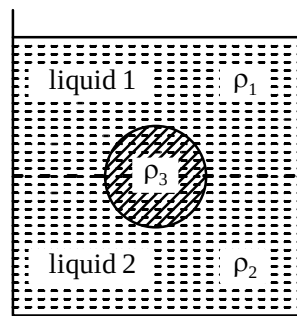
Roll Number : _____

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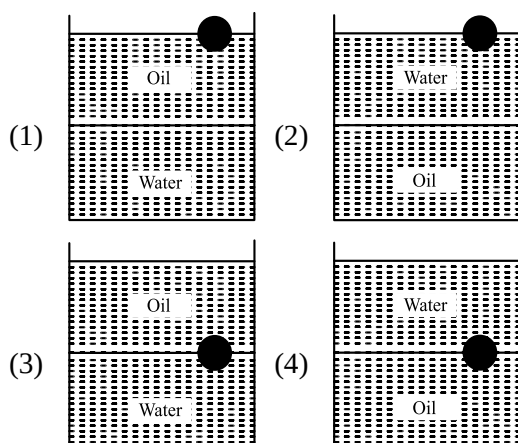
Candidate's Signature : _____ Invigilator's Signature _____

SECTION-(I) PHYSICS

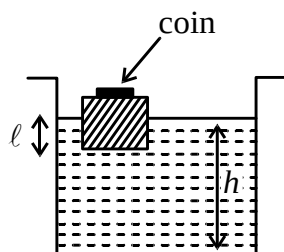
SECTION-A

- If the distance between two masses is doubled, the gravitational attraction between them:
 - is doubled.
 - becomes four times.
 - is reduced to half.
 - is reduced to a quarter.
- The distance of the centres of moon and earth is D . The mass of earth is 81 times the mass of the moon. At what distance from the centre of the earth, the gravitational force will be zero.
 - $\frac{D}{2}$
 - $\frac{2D}{3}$
 - $\frac{4D}{3}$
 - $\frac{9D}{10}$
- Mass M is divided into two parts xM and $(1-x)M$. For a given separation, the value of x for which the gravitational attraction between the two pieces becomes maximum is:
 - $\frac{1}{2}$
 - $\frac{3}{5}$
 - 1
 - 2
- Two spheres of mass m and M are situated in air and the gravitational force between them is F . The space around the masses is now filled with a liquid of specific gravity 3. The gravitational force will now be:
 - F
 - $\frac{F}{3}$
 - $\frac{F}{9}$
 - $3F$
- If the change in the value of ' g ' at a height h above the surface of the earth is the same as at a depth x below it, then (both x and h being much smaller than the radius of the earth)-
 - $x = h$
 - $x = 2h$
 - $x = \frac{h}{2}$
 - $x = h^2$
- Two planets have the same average density but their radii are R_1 and R_2 . If acceleration due to gravity on these planets be g_1 and g_2 respectively, then:
 - $\frac{g_1}{g_2} = \frac{R_1}{R_2}$
 - $\frac{g_1}{g_2} = \frac{R_2}{R_1}$
 - $\frac{g_1}{g_2} = \frac{R_1^2}{R_2^2}$
 - $\frac{g_1}{g_2} = \frac{R_1^3}{R_2^3}$
- When a body is taken from the equator to the poles, its weight?
 - Remains constant
 - Increases
 - Decreases
 - Increases at N -pole and decreases at S -pole
- If the earth suddenly shrinks (without changing mass) to half of its present radius, the acceleration due to gravity will be:
 - $g/2$
 - $4g$
 - $g/4$
 - $2g$
- A body of mass m rises to height $h = R/5$ from the earth's surface, where R is earth's radius. If g is acceleration due to gravity at earth's surface, the increase in potential energy is:
 - mgh
 - $\frac{4}{5}mgh$
 - $\frac{5}{6}mgh$
 - $\frac{6}{7}mgh$
- The flow speeds of air on the lower and upper surfaces of the wing of an aeroplane are v and $2v$ respectively. The density of air is ρ and surface area of wing is A . Find the dynamic lift on the wing.
 - $\frac{1}{2}\rho Av^2$
 - $\frac{3}{2}\rho Av^2$
 - $\frac{5}{2}\rho Av^2$
 - $2\rho Av^2$
- A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3 , is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1 , ρ_2 and ρ_3 ?
 
 - $\rho_1 < \rho_3 < \rho_2$
 - $\rho_3 < \rho_1 < \rho_2$
 - $\rho_1 > \rho_3 > \rho_2$
 - $\rho_1 < \rho_2 < \rho_3$

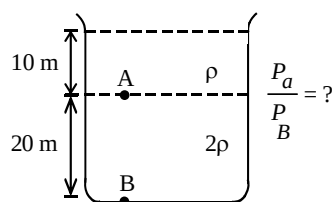
12. A ball is made of a material of density ρ where $\rho_{\text{oil}} < \rho < \rho_{\text{water}}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?



13. A wooden block, with a coin placed on its top, floats in water as shown in figure. The distance ℓ and h are shown here. After some time the coin falls into the water. Then



- (1) ℓ decreases and h increases
 (2) ℓ increase and h decreases
 (3) both ℓ and h increase
 (4) both ℓ and h decrease
14. For the given figure ratio of pressure at A and B will be ($\rho = 1000 \text{ kg/m}^3$, $P_0 = 10^5 \text{ Pa}$):



- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$
 (3) $\frac{1}{3}$ (4) $\frac{2}{3}$

15. A satellite with kinetic energy E_k is revolving round the earth in a circular orbit. How much more kinetic energy should be given to it so that it may just escape into outer space-

- (1) E_k (2) $2 E_k$
 (3) $\frac{1}{2} E_k$ (4) $3 E_k$

16. Earth is revolving around the sun if the distance of the Earth from the Sun is reduced to $\frac{1}{4}$ th of the present distance then the present day length reduced by-

- (1) $\frac{1}{4}$ (2) $\frac{1}{2}$
 (3) $\frac{1}{8}$ (4) $\frac{1}{6}$

17. The ratio of the lengths of two wires A and B of same material is 1 : 2 and the ratio of their diameter is 2 : 1. They are stretched by the same force, then the ratio of increase in length will be-

- (1) 2 : 1 (2) 1 : 4
 (3) 1 : 8 (4) 8 : 1

18. A wire is loaded by 6 kg at its one end, the increase in length is 12 mm. If the radius of the wire is doubled and all other magnitudes are unchanged, then increase in length will be:

- (1) 6 mm (2) 3 mm
 (3) 24 mm (4) 48 mm

19. If x longitudinal strain is produced in a wire of Young's modulus y , then energy stored in the material of the wire per unit volume is:

- (1) yx^2 (2) $2 yx^2$
 (3) $\frac{1}{2} y^2 x$ (4) $\frac{1}{2} yx^2$

20. In the Young's experiment, If length of wire and radius both are doubled then the value of Y will become:

- (1) 2 times (2) 4 times
 (3) Remains same (4) Half

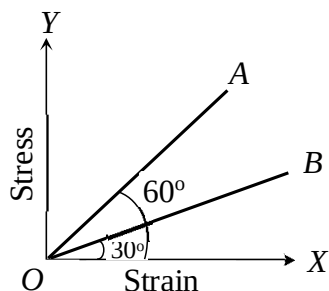
21. As we go down below the earth's surface, the acceleration due to gravity decreases by a factor ($d \rightarrow$ distance, $R \rightarrow$ radius of earth)

- (1) $1 + \frac{d}{R}$ (2) $1 - \frac{R}{d}$
 (3) $1 - \frac{d}{R}$ (4) Remains constant

22. If a rubber ball is taken at the depth of 200 m in a pool, its volume decreases by 0.1%. If the density of the water is $1 \times 10^3 \text{ kg/m}^3$ and $g = 10 \text{ m/s}^2$, then the volume elasticity in N/m^2 will be:

(1) 10^8
 (2) 2×10^8
 (3) 10^9
 (4) 2×10^9

23. The stress versus strain graphs for wires of two materials A and B are as shown in the figure. If Y_A and Y_B are the Young's modulus of the materials, then:



(1) $Y_B = 2Y_A$ (2) $Y_A = Y_B$
 (3) $Y_B = 3Y_A$ (4) $Y_A = 3Y_B$

24. Two particles of equal mass go round a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is:

(1) $v = \frac{1}{2R} \sqrt{\frac{1}{Gm}}$ (2) $v = \sqrt{\frac{Gm}{2R}}$
 (3) $v = \frac{1}{2} \sqrt{\frac{Gm}{R}}$ (4) $v = \sqrt{\frac{4Gm}{R}}$

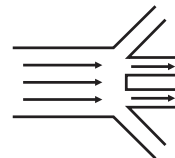
25. The radii of two planets are respectively R_1 and R_2 and their densities are respectively ρ_1 and ρ_2 . The ratio of the accelerations due to gravity at their surfaces is:

(1) $g_1 : g_2 = \frac{\rho_1}{R_1^2} : \frac{\rho_2}{R_2^2}$
 (2) $g_1 : g_2 = R_1 R_2 : \rho_1 \rho_2$
 (3) $g_1 : g_2 = R_1 \rho_2 : R_2 \rho_1$
 (4) $g_1 : g_2 = R_1 \rho_1 : R_2 \rho_2$

26. The escape velocity for a rocket from earth is 11.2 km/sec. Its value on a planet where acceleration due to gravity is double that on the earth and diameter of the planet is twice that of earth will be in km/sec:

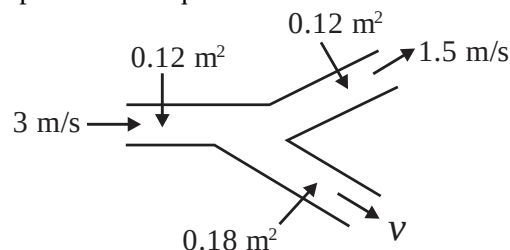
(1) 11.2
 (2) 5.6
 (3) 22.4
 (4) 53.6

27. Water is flowing through a channel that is 12 m wide with a speed of 0.75 m/s. The water then flows into four identical channels that have a width of 4.0 m. The density of the water does not change as it flows into the four channels. What is speed of the water in one of the smaller channels?



(1) 0.56 m/s (2) 2.3 m/s
 (3) 0.25 m/s (4) 0.75 m/s

28. An incompressible liquid flow, as shown. The speed of the liquid in the lower branch will be:

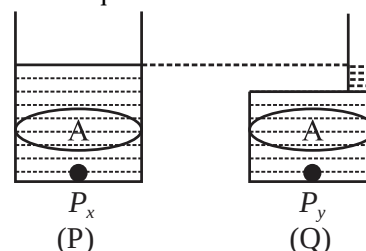


(1) 1.0 m/s (2) 1.5 m/s
 (3) 2.5 m/s (4) 3.0 m/s

29. Water is flowing through a horizontal pipe of the non-uniform cross-section. At the extreme narrow portion of the pipe, the water will have:

(1) Maximum speed and least pressure
 (2) Maximum pressure and least speed
 (3) Both pressure and speed maximum
 (4) Both pressure and speed minimum

30. Figure shows two containers P and Q with same base area A and each filled upto same height with same liquid. Select the **correct** alternative.

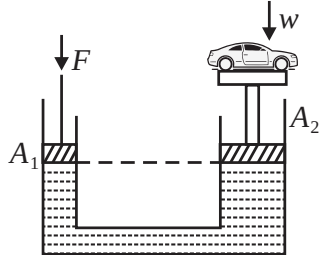


(1) $p_x = p_y$ (2) $p_x > p_y$
 (3) $p_y > p_x$ (4) Cannot say

31. A beaker containing a liquid of density ρ moves up with an acceleration 'a'. The pressure due to the liquid at a depth h below free surface of the liquid is:

(1) $h\rho g$ (2) $h\rho(g - a)$
 (3) $h\rho(g + a)$ (4) $2h\rho g \left(\frac{g - a}{g + a} \right)$

32. In a hydraulic jack as shown, mass of the car $W = 800 \text{ kg}$, $A_1 = 10 \text{ cm}^2$, $A_2 = 10 \text{ m}^2$. The minimum force F required to lift the car is:



- (1) 1 N
(2) 0.8 N
(3) 8 N
(4) 16 N
33. The Bulk modulus for an incompressible liquid is:
(1) Zero
(2) Unity
(3) Infinity
(4) Between 0 to 1
34. A boat carrying a number of stones is floating in a water tank. If the stones are unloaded into water, the water level in the tank will:
(1) remain unchanged.
(2) rise.
(3) fall.
(4) rise or fall depends on the number of stones unloaded.
35. Two liquids having densities d_1 and d_2 are mixed in such a way that both have same mass. The density of the mixture is:
(1) $\frac{d_1 + d_2}{2}$ (2) $\frac{d_1 + d_2}{d_1 d_2}$
(3) $\frac{d_1 d_2}{d_1 + d_2}$ (4) $\frac{2d_1 d_2}{d_1 + d_2}$

SECTION - B

36. Water ($\rho = 1000 \text{ kg/m}^3$) and kerosene ($\sigma = 800 \text{ kg/m}^3$) are filled in two identical cylindrical vessels. Both vessels have small holes at their bottom. The speed of the water and kerosene coming out of their holes are v_1 and v_2 respectively. Select the **correct** alternative:
(1) $v_1 = v_2$
(2) $v_1 = 0.8 v_2$
(3) $0.8 v_1 = v_2$
(4) $v_1 = \sqrt{0.8} v_2$

37. A large vessel of height H , is filled with a liquid of density ρ upto the brim. A small hole of radius r is made at the side vertical face, close to the base. The horizontal force is required to stop the gushing of liquid is:

- (1) $(\rho g H) \pi r^2$
(2) $\rho g H$
(3) $\rho g H \pi r$
(4) $\rho g \pi r^2$

38. A vertical U-tube of uniform cross-section contains water in the arms. A 10 cm glycerine column (R.D.=1.2) is added to one of the limbs. The level difference between the two free surfaces in the two limbs will be:

- (1) 4 cm (2) 2 cm
(3) 6 cm (4) 8 cm

39. The pressure at the bottom of a water tank is $4P$, P is atmospheric pressure. If water is drawn out till the water level decreases by $\frac{3}{5}$ th, then pressure at the bottom of the tank is:

- (1) $\frac{3P}{8}$ (2) $\frac{7P}{6}$
(3) $\frac{11P}{5}$ (4) $\frac{9P}{4}$

40. The gravitational field due to a mass distribution is $E = K/x^3$ in the x -direction. (K is a constant). Taking the gravitational potential to be zero at infinity, its value at a distance x is:

- (1) K/x (2) $K/2x$
(3) K/x^2 (4) $K/2x^2$

41. Radius of orbit of satellite of earth is R . Its kinetic energy is proportional to:

- (1) $\frac{1}{R}$
(2) $\frac{1}{\sqrt{R}}$
(3) R
(4) $\frac{1}{R^{3/2}}$

42. v_e and v_p denotes the escape velocity from the earth and another planet having twice the radius and the same mean density as the earth. Then

- (1) $v_e = v_p$ (2) $v_e = v_p/2$
(3) $v_e = 2v_p$ (4) $v_e = v_p/4$

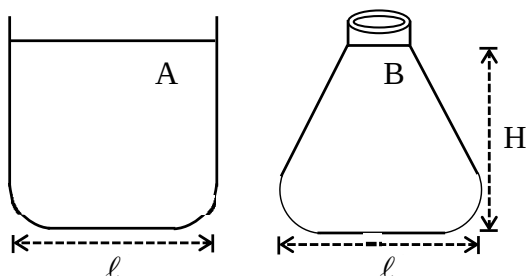
43. Select the **correct** statement from the following:

- (1) The orbital velocity of a satellite increases with the radius of the orbit
- (2) Escape velocity of a particle from the surface of the earth depends on the speed with which it is fired
- (3) The time period of a satellite does not depend on the radius of the orbit
- (4) The orbital velocity is inversely proportional to the square root of the radius of the orbit

44. For a satellite escape velocity is 11 km/s. If the satellite is launched at an angle of 60° with the vertical, then escape velocity will be:

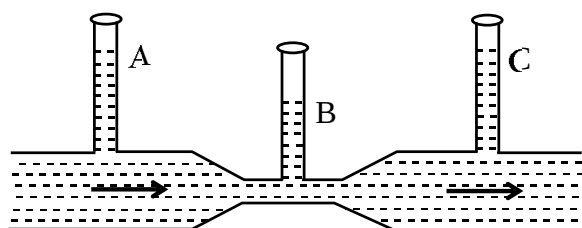
- (1) 11 km/s
- (2) $11\sqrt{3}$ km/s
- (3) $\frac{11}{\sqrt{3}}$ km/s
- (4) 33 km/s

45. Two vessels A and B have the same base area and contain water to the same height, but the mass of water in A is four times that in B. The ratio of the liquid thrust at the base of A to that at the base of B is:



- (1) 4 : 1
- (2) 2 : 1
- (3) 1 : 1
- (4) 16 : 1

46. Three tubes A, B and C are connected to a horizontal pipe in which ideal liquid is flowing. The radii of the tubes A, B and C at the junction are 2 cm, 1 cm and 2 cm respectively. It can be said:



- (1) The height of the liquid in the tube A is maximum
- (2) Height of the liquid in the tubes A and B is same
- (3) Height of the liquid in the tubes A, B and C is same
- (4) Height of the liquid in the tubes A and C is same

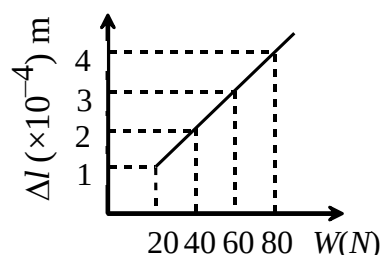
47. The orbital angular momentum of a satellite revolving at a distance r from the centre is L . If the distance is increased to $16r$, then the new angular momentum will be:

- (1) $16 L$
- (2) $64 L$
- (3) $\frac{L}{4}$
- (4) $4 L$

48. The Young's modulus of a wire is Y . If the energy per unit volume is E , then the strain will be:

- (1) $\sqrt{\frac{2E}{Y}}$
- (2) $\sqrt{2EY}$
- (3) EY
- (4) $\frac{E}{Y}$

49. The adjacent graph shows the extension (Δl) of a wire of length 1m suspended from the top of a roof at one end with a load W connected to the other end. If the cross-sectional area of the wire is 10^{-6} m^2 , calculate the young's modulus of the material of the wire.



- (1) $2 \times 10^{11} \text{ N/m}^2$
- (2) $2 \times 10^{-11} \text{ N/m}^2$
- (3) $3 \times 10^{-12} \text{ N/m}^2$
- (4) $3 \times 10^{-13} \text{ N/m}^2$

50. A tank of height 5 m is full of water. There is a hole of cross sectional area 10 cm^2 in its bottom. The initial rate of volume of water that will come out from this hole is ($g = 10 \text{ m/s}^2$):

- (1) $10^{-3} \text{ m}^3/\text{s}$
- (2) $10^{-4} \text{ m}^3/\text{s}$
- (3) $10 \text{ m}^3/\text{s}$
- (4) $10^{-2} \text{ m}^3/\text{s}$

SECTION-(II) CHEMISTRY

SECTION - A

51. If the concentration of the reactants is increased, the rate of reaction:
- (1) always remains unaffected.
 - (2) increases.
 - (3) decreases.
 - (4) may increase or decrease.
52. The rate of reaction depends upon the:
- (1) Temperature
 - (2) concentration of reactant.
 - (3) pressure (for gaseous reaction).
 - (4) All of these
53. In the reaction $A + 2B \rightarrow 6C + 2D$ if rate of disappearance of A is $2.6 \times 10^{-2} \text{ Ms}^{-1}$, what will be the value of rate of disappearance of B?
- (1) $8.5 \times 10^{-2} \text{ Ms}^{-1}$
 - (2) $2.6 \times 10^{-2} \text{ Ms}^{-1}$
 - (3) $5.2 \times 10^{-2} \text{ Ms}^{-1}$
 - (4) $7.5 \times 10^{-2} \text{ Ms}^{-1}$
54. Electrolytic conduction differs from metallic conduction is that in the case of electrolytic conduction:
- (1) the resistance increases with increasing temperature.
 - (2) the resistance decreases with increasing temperature.
 - (3) The flow of current does not generate heat.
 - (4) the resistance is independent of the length of the conductor.
55. Given $\ell / A = 0.5 \text{ cm}^{-1}$, $R = 50 \text{ ohm}$, $N = 1.0$, then equivalent conductance of the electrolytic cell is:
- (1) $10 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$
 - (2) $20 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$
 - (3) $300 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$
 - (4) $100 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$
56. The increase in equivalent conductance of an electrolyte solution with dilution is due to the increase in:
- (1) ionic attraction.
 - (2) molecular attraction.
 - (3) degree of association of the electrolyte
 - (4) degree of ionisation of the electrolyte.

57. The molar conductivity is maximum for the solution of concentration:
- (1) 0.001 M
 - (2) 0.005 M
 - (3) 0.002 M
 - (4) 0.004 M
58. Which one of the following statements for the order of a reaction is **incorrect**?
- (1) Order can be determined only experimentally.
 - (2) Order is not influenced by stoichiometric coefficient of the reactants.
 - (3) Order of reaction is sum of power to the concentration terms of reactants to express the rate of reaction.
 - (4) Order of reaction is always whole number.
59. The half-life of a substance in a certain enzyme-catalysed first order reaction is 138 s. The time required for the concentration of the substance to fall from 1.28 mg L^{-1} to 0.04 mg L^{-1} :
- (1) 276 s
 - (2) 414 s
 - (3) 552 s
 - (4) 690 s
60. Rutherford's α -particle scattering experiments led to the conclusion that:
- (1) mass and energy related.
 - (2) the mass and the positive charge of an atom are concentrated in the nucleus.
 - (3) neutrons are present in the nucleus.
 - (4) atoms are electrically neutral.

61. Match List-I with List-II.

List-I		List-II	
(A)	${}_{26}\text{Fe}^{54}, {}_{26}\text{Fe}^{56}, {}_{26}\text{Fe}^{57}, {}_{26}\text{Fe}^{58}$	(I)	Isotopes
(B)	${}_1\text{H}^3, {}_2\text{He}^3$	(II)	Isotones
(C)	${}_{32}\text{Ge}^{76}, {}_{33}\text{As}^{77}$	(III)	Isodiaphers
(D)	${}_{92}\text{U}^{235}, {}_{90}\text{Th}^{231}$	(IV)	Isobars

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
 - (2) A-IV, B-III, C-II, D-I
 - (3) A-II, B-III, C-IV, D-I
 - (4) A-II, B-I, C-III, D-IV
62. Calculate wave number in cm^{-1} of radiation have wavelength of $4600 \text{ \AA}$:
- (1) 15440 cm^{-1}
 - (2) 21739 cm^{-1}
 - (3) 40000 cm^{-1}
 - (4) 50507 cm^{-1}

63. The rate constant of the reaction $A \rightarrow B$ is $0.6 \times 10^{-3} \text{ M sec}^{-1}$. If the concentration of A is 5 M, then concentration of B after 20 min is:
- (1) 0.36 M (2) 0.72 M
(3) 1.08 M (4) 3.60 M

64. Which of the following is **not** possible set of quantum number for a particular electron in an atom is:

- (1) $n = 3, l = 2, m = -2, s = -1/2$
(2) $n = 4, l = 0, m = 0, s = -1/2$
(3) $n = 5, l = 3, m = 0, s = +1/2$
(4) $n = 3, l = 2, m = -3, s = -1/2$

65. The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be:

- (1) $\lambda_1 = \lambda_2$
(2) $\lambda_1 = 2\lambda_2$
(3) $\lambda_1 = 4\lambda_2$
(4) $\lambda_1 = 3\lambda_2$

66. Match List-I with List-II.

List-I		List-II	
(A)	$t_{1/2} = \text{Constant}$	(I)	First-order
(B)	$t_{1/2} \propto a$	(II)	Third-order
(C)	$t_{1/2} \propto 1/a$	(III)	Second-order
(D)	$t_{1/2} \propto 1/a^2$	(IV)	Zero-order

a = initial concentration of the reactant

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-III, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

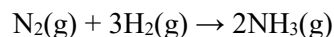
67. The value of Planck's constant is $6.63 \times 10^{-34} \text{ Js}$. The speed of light is $3 \times 10^{17} \text{ nm s}^{-1}$. Which value is closest to the wavelength in nano-meter of a quantum of light with frequency of $6 \times 10^{15} \text{ s}^{-1}$?

- (1) 75 nm
(2) 10 nm
(3) 25 nm
(4) 50 nm

68. Specific conductance of 0.1 M nitric acid is $6.3 \times 10^{-2} \text{ ohm}^{-1} \text{ cm}^{-1}$. The molar conductance of solution is:

- (1) $630 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
(2) $315 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
(3) $100 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
(4) $6300 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$

69. Consider the chemical reaction:



The rate of this reaction can be expressed in terms of concentration of $\text{N}_2(\text{g})$, $\text{H}_2(\text{g})$ or $\text{NH}_3(\text{g})$. Identify the **correct** relationship amongst the rate expressions.

- (1) Rate of reaction $= -\frac{d[\text{N}_2]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$
(2) Rate of reaction $= \frac{d[\text{N}_2]}{dt} = -\frac{3d[\text{H}_2]}{dt} = \frac{2d[\text{NH}_3]}{dt}$
(3) Rate of reaction $= \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$
(4) Rate of reaction $= \frac{d[\text{N}_2]}{dt} = \frac{d[\text{H}_2]}{dt} = \frac{d[\text{NH}_3]}{dt}$

70. For the reaction $A + B \rightarrow \text{products}$, it is observed that:

- (a) on doubled the initial concentration of A only, the rate of reaction is also doubled and
(b) on doubling the initial concentrations of both A and B, there is a change by a factor of 8 in the rate of the reaction.

The rate of this reaction is given by:

- (1) $\text{rate} = k[\text{A}][\text{B}]$
(2) $\text{rate} = k[\text{A}]^2[\text{B}]$
(3) $\text{rate} = k[\text{A}][\text{B}]^2$
(4) $\text{rate} = k[\text{A}]^2[\text{B}]^2$

71. Half-life period of a first-order reaction is 1386 s. The specific rate constant of the reaction is:

- (1) $5.0 \times 10^{-2} \text{ s}^{-1}$ (2) $5.0 \times 10^{-3} \text{ s}^{-1}$
(3) $0.5 \times 10^{-2} \text{ s}^{-1}$ (4) $0.5 \times 10^{-3} \text{ s}^{-1}$

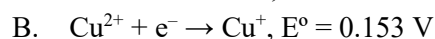
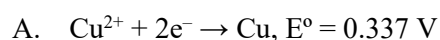
72. For an endothermic reaction, energy of activation is E_a and enthalpy of reaction is ΔH (both of these in kJ/mol). Minimum value of forward E_a will be:

- (1) equal to zero. (2) less than ΔH .
(3) equal to ΔH . (4) more than ΔH .

73. The specific conductance of a 0.1 N KCl solution at 23°C is $0.012 \text{ ohm}^{-1} \text{ cm}^{-1}$. The resistance of cell containing the solution at the same temperature was found to be 55 ohm. The cell constant will be:

- (1) 0.142 cm^{-1} (2) 0.66 cm^{-1}
(3) 0.918 cm^{-1} (4) 1.12 cm^{-1}

74. Given:



Electrode potential, E° for the reaction, $\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$, will be:

- (1) 0.38 V (2) 0.52 V
(3) 0.90 V (4) 0.30 V

75. Calculate the radius ratio of 2nd and 4th orbit of Li²⁺:

- (1) 1 : 2 (2) 2 : 5
(3) 1 : 4 (4) 3 : 4

76. Angular momentum for p-subshell electron:

- (1) $\frac{3h}{\pi}$
(2) zero
(3) $\frac{\sqrt{2}h}{2\pi}$
(4) None of these

77. How much time an e⁻ will take for one complete revolution in 2nd excited state of Li²⁺?

- (1) 2.56×10^{-16} s
(2) 4.5×10^{-16} s
(3) 8.59×10^{-16} s
(4) 3.35×10^{-16} s

78. The charge on the atom containing 17 protons, 18 neutrons and 18 electrons is:

- (1) +1
(2) -2
(3) -1
(4) zero

79. For a cell involving one electron $E_{\text{cell}}^{\circ} = 0.59$ V at 298 K, the equilibrium constant for the cell reaction is:

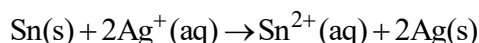
$$\left[\text{Given that } \frac{2.303RT}{F} = 0.059 \text{ V at } T = 298 \text{ K} \right]$$

- (1) 1.0×10^2
(2) 1.0×10^5
(3) 1.0×10^{10}
(4) 1.0×10^{30}

80. The **correct** order of chemical reactivity with water according to electrochemical series:

- (1) K > Mg > Zn > Cu
(2) Mg > Zn > Cu > K
(3) K > Zn > Mg > Cu
(4) Cu > Zn > Mg > K

81. Which will increase the voltage of the cell



- (1) increase in the concentration of Ag⁺ ions.
(2) increase in the concentration of Sn²⁺ ions.
(3) increase in size of the silver rod.
(4) None of these

82. If mass of neutron is halved and mass of proton is doubled and mass of electron is made 4 times then find out the new atomic mass of $^{16}_8\text{O}$ and the percent by which it is increased:

- (1) 50%
(2) 40%
(3) 25%
(4) 60%

83. Activation energy (E_a) and rate constants (K_1 and K_2) of a chemical reaction at two different temperature (T_1 and T_2) are related by:

- (1) $\ln \frac{K_2}{K_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} + \frac{1}{T_1} \right)$
(2) $\ln \frac{K_2}{K_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$
(3) $\ln \frac{K_2}{K_1} = -\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$
(4) $\ln \frac{K_1}{K_2} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$

84. A reaction having equal energies of activation for forward and reverse reactions has:

- (1) $\Delta H = \Delta G = \Delta S = 0$
(2) $\Delta S = 0$
(3) $\Delta G = 0$
(4) $\Delta H = 0$

85. When initial concentration of a reactant is doubled in a reaction, its half-life period is **not** affected. The order of the reaction is:

- (1) first
(2) second
(3) more than zero but less than first.
(4) zero

SECTION - B

86. Electromagnetic radiation travels through vacuum at speed of:

- (1) 186000 ms^{-1}
(2) 125 ms^{-1}
(3) $3.00 \times 10^8 \text{ ms}^{-1}$
(4) it depends upon wavelength.

87. Calculate the wavelength of a photon having an energy of 5 eV:

- (1) $4.5 \times 10^{-7} \text{ m}$ (2) $2.5 \times 10^{-7} \text{ m}$
(3) $3.85 \times 10^{-7} \text{ m}$ (4) $6.8 \times 10^{-7} \text{ m}$

88. Consider the following relations for emf of an electrochemical cell:
- emf of cell = (Oxidation potential of anode) – (Reduction potential of cathode)
 - emf of cell = (Oxidation potential of anode) + (Reduction potential of cathode)
 - emf of cell = (Reduction potential of anode) + (Reduction potential of cathode)
 - emf of cell = (Oxidation potential of anode) – (Oxidation potential of cathode)
- Which of the above relation are **correct**:
- C and B
 - C and D
 - B and D
 - C and A
89. If the E_{cell}° for a given relation has a negative value, then which of the following gives the **correct** relationship for the value of ΔG° and K_{eq} ?
- $\Delta G^{\circ} > 0$; $K_{\text{eq}} > 1$
 - $\Delta G^{\circ} < 0$; $K_{\text{eq}} > 1$
 - $\Delta G^{\circ} < 0$; $K_{\text{eq}} < 1$
 - $\Delta G^{\circ} > 0$; $K_{\text{eq}} < 1$
90. Molar conductivities (Λ_m°) at infinite dilution of NaCl, HCl and CH_3COONa are 126.4, 425.9 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. Λ_m° for CH_3COOH will be:
- $290.8 \text{ S cm}^2 \text{ mol}^{-1}$
 - $390.5 \text{ S cm}^2 \text{ mol}^{-1}$
 - $425.5 \text{ S cm}^2 \text{ mol}^{-1}$
 - $180.5 \text{ S cm}^2 \text{ mol}^{-1}$
91. During the electrolysis of molten sodium chloride, the time required to produce 0.10 mole of chlorine gas using a current of 3 A is:
- 220 minutes
 - 330 minutes
 - 55 minutes
 - 107 minutes
92. When an electron in H-atom jumps from $n = 4$ to $n = 1$, ultra violet light is emitted, if the transition corresponds from $n = 4$ to $n = 2$, then electron falls in which of the following region?
- Ultra violet region
 - Visible region
 - Infrared region
 - None of these
93. The uncertainty in position of an electron and helium atom are same. If the uncertainty in momentum for the electron is $32 \times 10^5 \text{ kg ms}^{-1}$ then the uncertainty in momentum of helium atom will be:
- $32 \times 10^5 \text{ kg ms}^{-1}$
 - $16 \times 10^5 \text{ kg ms}^{-1}$
 - $8 \times 10^5 \text{ kg ms}^{-1}$
 - None of these
94. n and ℓ values of an orbital 'A' are 3 and 2, and another orbital 'B' are 5 and 0. The energy of:
- B is more than A.
 - A is more than B.
 - A and B are of same energy.
 - None of these
95. Maximum number of electrons in a subshell of an atom is determined by following:
- $2n^2$
 - $4\ell + 2$
 - $2\ell + 1$
 - $4\ell - 2$
96. A first order reaction has a rate constant of $2.303 \times 10^{-3} \text{ s}^{-1}$. The time required for 40 g of this reactant to reduce to 10 g will be:
- 230.3 s
 - 301 s
 - 2000 s
 - 602 s
97. Given below are two statements:
- Statement I:** In zero-order reactions, half-life is half of the completion time of reaction.
- Statement II:** Half-life is half of the time for completion of reaction of any order.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
98. Which of the following is strong electrolyte.
- NH_4OH
 - CH_3COOH
 - HCN
 - HCl
99. Two electrons occupying the same orbital are distinguished by:
- principal quantum number.
 - magnetic quantum number.
 - azimuthal quantum number.
 - spin quantum number.
100. Orbital having 3 angular nodes and 3 total nodes is:
- 5p
 - 3d
 - 4f
 - 6d

SECTION-(III) BOTANY

SECTION - A

- 101.** Identify the **correct** statement w.r.t to blue-green algae.
- (1) Possess mucilaginous sheath.
 - (2) Absence of rigid cell wall.
 - (3) Contain chlorophyll *c* only.
 - (4) Found in marine habitat only.
- 102.** Acellular organisms which are **not** put in any kingdom by Whittaker are:
- (1) viruses and lichens.
 - (2) viruses and viroids.
 - (3) protists, viroids and prions.
 - (4) viruses, lichens and mycorrhiza.
- 103.** Select the **incorrect** statement regarding fungi.
- (1) Ascomycetes are commonly known as sac-fungi.
 - (2) *Penicillium* is unicellular and are source of antibiotics.
 - (3) *Neurospora* is used extensively in biochemical and genetic work.
 - (4) Morels and truffles are edible fungi.
- 104.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): In the five kingdom classification bacteria are included in kingdom Monera.
Reason (R): Bacteria show the least metabolic diversity.
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is NOT the correct explanation of A.
- 105.** Select the **wrong** statement.
- (1) *Paramoecium* have a cavity (gullet) that opens to the outside of the cell surface.
 - (2) Spores of slime moulds are dispersed by water currents.
 - (3) All protozoans are heterotrophs and live as predators or parasites.
 - (4) Amoeboid protozoans move and capture their prey by putting out pseudopodia.
- 106.** Euglenoids have:
- (1) two flagella of same size.
 - (2) lipid rich layer called pellicle.
 - (3) pigments identical to those present in higher plants.
 - (4) heterotrophic mode of nutrition only.
- 107.** Read the following and identify the **incorrect** statements.
- A. Virus means venom or poisonous fluid.
 - B. A virus is a nucleoprotein and the genetic material is infectious.
 - C. In general, viruses that infect plants have double stranded RNA.
 - D. A virus can contain both RNA and DNA.
- Choose the most appropriate answer from the options given below.
- (1) A and B only
 - (2) B and C only
 - (3) C and D only
 - (4) A and D only
- 108.** Heterocyst present in *Anabaena* is specialised for:
- (1) production of antibiotics.
 - (2) fission.
 - (3) nitrogen fixation.
 - (4) sexual reproduction.
- 109.** Diatoms and golden algae are found in:
- (1) mostly terrestrial environment.
 - (2) marine environments only.
 - (3) fresh water only.
 - (4) freshwater as well as in marine environments.
- 110.** Spores of slime moulds possess **A** and survive for many years under **B**. Identify **A** and **B** are respectively.
- | A | B |
|-------------------------|---------------------|
| (1) true walls | suitable conditions |
| (2) non cellulosic wall | adverse conditions |
| (3) true walls | adverse conditions |
| (4) non cellulosic wall | suitable conditions |
- 111.** The members of deuteromycetes are commonly known as imperfect fungi because:
- (1) they show only asexual or vegetative phases.
 - (2) they produce meiospores only.
 - (3) they have aseptate mycelium.
 - (4) they lack cell wall.

- 112.** Identify the **correct** statement w.r.t archaeobacteria.
- (1) They are not capable of tolerating high temperature.
 - (2) They can present in the gut of several ruminant animals.
 - (3) They never found in extreme saline environment.
 - (4) They are not used for the production of biogas.

- 113.** Given below are two statements:

Statement I: The cell walls of fungi are composed of chitin and polysaccharides.

Statement II: Most fungi are saprophytic in their mode of nutrition.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

- 114.** Select the **correct** statement w.r.t characteristic features of the kingdom.

- (1) In animalia, the mode of nutrition is autotrophic.
- (2) In monera, the nuclear membrane is absent.
- (3) In fungi, the lichens are included.
- (4) In plantae, prokaryotic chlorophyll containing organism are included.

- 115.** Both *Neurospora* and *Penicillium* form:

- (1) endogenous asexual and exogenous sexual spores.
- (2) conidia as asexual and basidiospore as sexual spores.
- (3) zoospore as asexual and ascospore as sexual spores.
- (4) exogenous asexual and endogenous sexual spores.

- 116.** Identify the group that includes cyanobacteria.

- (1) Chemosynthetic autotrophic bacteria
- (2) Photosynthetic autotrophic bacteria
- (3) Photosynthetic chrysophytes.
- (4) Heterotrophic bacteria.

- 117.** Members of basidiomycetes do **not** have:

- (1) fruiting bodies.
- (2) dikaryophase in their life cycle.
- (3) sexual spores.
- (4) sex organs.

- 118.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Boundaries of Kingdom Protista are not well defined.

Reason R: The protistan cell body contains a well defined nucleus and other membrane-bound organelles.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

- 119.** Which of the following often forms bloom in polluted water bodies?

- (1) Protozoans
- (2) Archaeobacteria
- (3) Slime moulds
- (4) Cyanobacteria

- 120.** Which of the following is **not** a characteristic of Protista?

- (1) Some have flagella or cilia.
- (2) Primarily aquatic.
- (3) Includes multicellular eukaryotes.
- (4) Reproduce asexually as well as sexually.

- 121.** Given below are two statements:

Statement I: Heterotrophic bacteria have a significant impact on human affairs.

Statement II: Chemosynthetic-autotrophic bacteria play a great role in recycling nutrients like nitrogen, phosphorous and sulphur.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

122. An infectious spore-like stage is present in the life cycle of a protozoan which causes:
- (1) sleeping sickness.
 - (2) mumps.
 - (3) polio.
 - (4) malaria.

123. Bacteriophage is a:
- (1) bacteria that infects fungi.
 - (2) bacteria that infects virus.
 - (3) virus that infects fungi.
 - (4) virus that infects bacteria.

124. Mushrooms, bracket fungi or puffballs are commonly known forms of;
- (1) phycomycetes.
 - (2) ascomycetes.
 - (3) basidiomycetes.
 - (4) deuteromycetes.

125. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Dinoflagellates	(I)	Cell wall contain silica
(B)	Protozoans	(II)	In absence of sunlight they behave like heterotrophs
(C)	Diatoms	(III)	Primitive relatives of animals
(D)	Euglenoids	(IV)	Releases toxins that kill marine animals

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
 - (2) A-IV, B-I, C-II, D-III
 - (3) A-III, B-I, C-II, D-IV
 - (4) A-IV, B-III, C-I, D-II
126. Under suitable conditions, slime moulds form an aggregation called_____.
- (1) plasmodium
 - (2) conidia
 - (3) fruiting bodies
 - (4) pellicle
127. Which of the following statement is **incorrect** regarding mycoplasma ?
- A. Has no cell wall
 - B. Is the smallest living organism known
 - C. Cannot survive without oxygen
 - D. Are pathogenic in animals and plants
 - E. Are archaebacteria
- Choose the most appropriate answer from the options given below:
- (1) C and E only
 - (2) C and D only
 - (3) A, B and D only
 - (4) A, C and D only

128. Identify the **incorrect** statement w.r.t diatoms?

- (1) Most of them are predators.
- (2) Their cell walls are indestructible.
- (3) The cell wall is like a soap box.
- (4) They are microscopic and float passively in water currents.

129. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Slime moulds are saprophytic protists.

Reason R: Slime moulds body moves along decaying twigs and leaves engulfing organic material.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

130. Fusion of protoplasts between two motile or non-motile gametes is called as:

- (1) plasmogamy.
- (2) karyogamy.
- (3) dikaryon.
- (4) dikaryophase.

131. Identify the **correct** option for red tide causing organisms?

- (1) Having one flagellum
- (2) Fresh water organism
- (3) Rapid multiplication
- (4) Are the chief producers in the ocean

132. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Aspergillus</i>	(I)	Zygospore
(B)	<i>Mucor</i>	(II)	Imperfect fungi
(C)	<i>Puccinia</i>	(III)	Ascospore
(D)	<i>Trichoderma</i>	(IV)	Basidiospore

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-IV, C-III, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-I, C-IV, D-II

133. How many of the given organisms have coenocytic mycelium?

Bracket fungi, *Neurospora*, *Rhizopus*, Smut fungi, *Trichoderma*, *Albugo*, *Alternaria*, *Mucor*, Mushroom

- (1) Five
- (2) Three
- (3) Four
- (4) Two

134. Which of the following statement is **incorrect** regarding the two kingdom system of classification?

- (1) Two kingdom system of classification was given by Linnaeus.
- (2) Two kingdom system of classification includes Plantae and Animalia.
- (3) This system distinguishes between photosynthetic and non-photosynthetic organisms.
- (4) This system did not distinguish between unicellular and multicellular organisms.

135. Which of the following option **correctly** exemplifies the organism with its associated group?

- (1) *Alternaria* belongs to algal fungi
- (2) *Agaricus* belongs to imperfect fungi
- (3) *Albugo* belongs to sac fungi
- (4) *Colletotrichum* belongs to imperfect fungi

SECTION - B

136. Read the following statements about viruses.

- A. Viruses are obligate intracellular parasites and remain inert outside to their specific host cell.
- B. Viruses are not considered truly living.
- C. The RNA of the viroid was of low molecular weight.
- D. Prions are larger in size to viruses.

How many of the following statement(s) is/are **correct**?

- (1) Four
- (2) Three
- (3) Two
- (4) One

137. The fungi which are decomposers of litter and help in mineral cycling belong to:

- (1) ascomycetes.
- (2) basidiomycetes.
- (3) deuteromycetes.
- (4) phycomycetes.

138. All of the following are included in Kingdom Protista, **except**:

- (1) protozoans.
- (2) chrysophytes.
- (3) slime moulds.
- (4) bladderwort.

139. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The bacterial structure is very complex but they are very simple in behaviour.

Reason R: Many of bacteria live in or on other organisms as parasites.

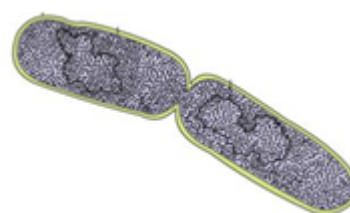
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

140. Sporozoans differ from ciliated protozoans because they:

- (1) use cilia for locomotion.
- (2) cause disease which has a staggering effect on human population.
- (3) use flagella for movement.
- (4) use pseudopodia for capturing prey.

141. Select the **incorrect** statement w.r.t to the below diagram.



- (1) It is reproduce mainly by fission.
- (2) Under unfavourable conditions, they produce spores.
- (3) Herpes and influenza are well known diseases caused by them.
- (4) They also reproduce by a sort of sexual reproduction.

142. Which of the following statement is **not** correct?

- (1) Eubacteria are true bacteria.
- (2) Cyanobacteria are eubacteria.
- (3) Chemosynthetic autotrophic bacteria oxidise inorganic substances.
- (4) Autotrophic bacteria are most abundant in nature.

143. Identify the **correct** statement(s) from the following.

- A. The cyanobacteria have chlorophyll *a* similar to green plants.
- B. Cyanobacteria are unicellular, colonial or filamentous.
- C. Archaeobacteria and eubacteria having a similar cell wall structure.
- D. Chemosynthetic autotrophic bacteria are helpful in making curd from milk.
- E. The majority of heterotrophic bacteria are important decomposers.

- (1) A, B and C
- (2) B, C and D
- (3) A, B and E
- (4) C, D and E

144. Identify the **correct** statement w.r.t basidiospores.

- (1) They are produced on basidium.
- (2) They are asexual spores.
- (3) Two basidiospores are produced by mitosis.
- (4) They are endogenously produced spores.

145. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Rhizopus</i>	(I)	Causes mad cow disease in cattle
(B)	Prions	(II)	Causes potato spindle tuber disease
(C)	Animalia	(III)	Organ system level of body organisation
(D)	Viroids	(IV)	Bread mould

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-III, D-II
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-IV, B-I, C-II, D-III

146. Which among the following is **not** the basis of classification of Kingdom Fungi into various classes?

- (1) Mode of spore formation
- (2) Mode of nutrition
- (3) Fruiting bodies
- (4) Morphology of mycelium

147. Given below are two statements:

Statement I: Lichens are symbiotic association between algae and fungi.

Statement II: In lichen, fungi prepare food for algae and algae provide shelter and absorb mineral nutrients and water for its partner.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

148. Infectious agent that causes potato spindle tuber disease:

- (1) is found to be a free DNA.
- (2) lacks the protein coat.
- (3) was discovered by Dimitri Ivanowsky.
- (4) is bigger than viruses.

149. Read the following and choose the **incorrect** statements.

- A. Venus fly trap is a parasitic plant.
- B. *Cuscuta* is an insectivorous plant.
- C. Plantae includes algae, bryophytes, pteridophytes, gymnosperms and angiosperms.
- D. Life cycle of plants has two distinct phases – the diploid sporophytic and the haploid gametophytic.

- (1) A and B only
- (2) B and C only
- (3) C and D only
- (4) A and D only

150. Both diatoms and dinoflagellates are protists but differ in:

- (1) mode of nutrition.
- (2) cell wall composition.
- (3) cell type.
- (4) habitat.

SECTION-(IV) ZOOLOGY

SECTION - A

151. The amount of filtrate formed by the kidney per minute is:

- (1) renal blood flow.
- (2) glomerular filtration rate.
- (3) urinary output.
- (4) renal plasma flow.

152. The filtration membrane consists of:

- (1) endothelium of glomerular blood vessels only.
- (2) epithelium of Bowman's capsule only.
- (3) a basement substance between these two layers only.
- (4) all of these

153. Match List-I with List-II.

List-I		List-II	
(A)	Ammonotelic	(I)	Renal tubule
(B)	Uricotelic	(II)	Reptiles
(C)	Ureotelic	(III)	Marine fishes
(D)	Bowman's capsule	(IV)	Many bony fishes

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-IV, C-II, D-I
- (4) A-IV, B-II, C-III, D-I

154. Antennal gland help in removal of nitrogenous waste in:

- (1) *Amphioxus*
- (2) Prawn
- (3) *Planaria*
- (4) Rotifers

155. Given below are two statements:

Statement I: Human kidneys can produce urine nearly four times concentrated than the initial filtrate formed.

Statement II: The functioning of the kidneys is monitored and regulated by hormonal feedback mechanisms involving the hypothalamus, JGA and heart.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect

156. The renal cortex extends between the medullary pyramid as;

- (1) renal pelvis.
- (2) renal corpuscle.
- (3) columns of Bertini.
- (4) calyx.

157. Match List-I with List-II.

List-I		List-II	
(A)	Flatworm	(I)	Excrete urea
(B)	Land snail	(II)	Excrete uric acid
(C)	<i>Periplaneta</i>	(III)	Flame cell
(D)	Mammals	(IV)	Malpighian tubule

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-II, C-IV, D-I
- (4) A-IV, B-II, C-III, D-I

158. Nearly all of the essential nutrients and 70-80 percent of electrolytes and water are reabsorbed in which segment of nephron?

- (1) PCT
- (2) Ascending loop of henle
- (3) Descending loop of henle
- (4) DCT

159. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

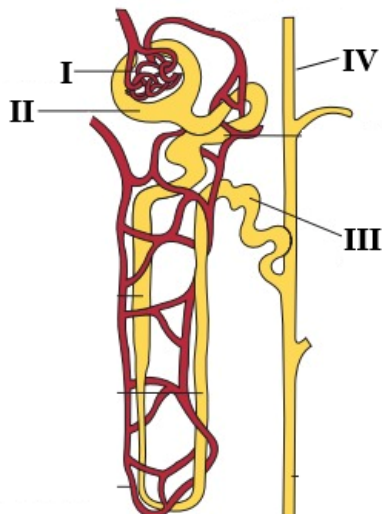
Assertion A: Tubular secretion is an important step in urine formation.

Reason R: During urine formation, the tubular cells secrete substances like H^+ , K^+ and ammonia into the filtrate.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

160. The given below diagram is the representation of a nephron with labelled I, II, III and IV. Identify the part where filtrate in nephron is called urine.



- (1) I
(2) II
(3) III
(4) IV
161. Presence of glucose in urine are indicative of;
(1) renal calculi.
(2) uremia.
(3) diabetes mellitus.
(4) diabetes insipidus.
162. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Ascending limb of loop of henle is permeable to water and allow transport of electrolyte actively and passively.
Reason R: As the concentrated filtrate pass upward, it gets diluted due to the passage of electrolytes to the medullary fluid.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.
163. Amount of CO₂ removed by our lungs per minute is;
(1) 200 ml.
(2) 300 ml.
(3) 100 ml.
(4) 500 ml.

164. Read the following statements (A-E) and select the **correct** option.
A. The excretory system consists of a pair of kidneys, one pair of ureters, a urinary bladder and a urethra.
B. Kidneys are situated between the levels of third thoracic and last third lumbar vertebra.
C. Average weight of human kidney is 120-170 g.
D. Hilum is a notch towards the centre of inner concave surface of kidney.
E. Inside the kidney, there are two zones, an outer cortex and an inner medulla.
Choose the most appropriate answer from the options given below:
(1) B, C, D, and E only.
(2) A, B and C only.
(3) B, D and E only.
(4) A, C, D and E only.
165. The filtration slit are formed by which of the following layers of filtration membrane?
(1) Endothelium of glomerular blood vessels.
(2) Podocyte of Bowman's capsule.
(3) Endothelium of vasa recta.
(4) Basement membrane of PCT.
166. Select the **incorrect** statement w.r.t cortical nephron.
(1) Loop of henle is short.
(2) They are the most abundant type of nephrons in kidney.
(3) Vasa recta is well developed in cortical nephrons.
(4) Malpighian corpuscle PCT and DCT of the nephrons are situated in cortex.
167. Which of the following will **not** cause micturition?
(1) Stretching of urinary bladder.
(2) Contraction of smooth muscle of bladder.
(3) Relaxation of muscles of sphincter.
(4) Activation of stretch receptor in the wall of ureters.
168. The medullary interstitial fluid has a very high osmolarity which is maintained by;
(1) movement of urea from collecting duct into medullary interstitium.
(2) movement of NaCl from descending limb of Henle loop into medullary interstitium.
(3) movement of K⁺ and Cl⁻ from PCT and DCT into peri-tubular capillaries.
(4) uric acid and H⁺ interstitium.

- 169.** The release of renin from JG Cell is triggered in response to;
- (1) fall in glomerular filtration rate.
 - (2) increase in blood pressure.
 - (3) release in ANF.
 - (4) release of aldosterone.
- 170.** Choose the **incorrect** statement w.r.t counter current mechanism.
- (1) Osmolarity of filtrate increase while moving down the descending limb of loop of Henle.
 - (2) There will be no urine formation if loop of Henle is absent in mammalian nephrons.
 - (3) NaCl and urea maintain osmolarity gradient in medulla.
 - (4) Osmolarity of fluid moving out from the collecting duct is four times that of plasma.
- 171.** Read the following statements (A-D) and select the **correct** statement.
- A. Bowman's capsule is double walled cup-like structure.
 - B. Glomerulus along with Bowman's capsule is called the malpighian body.
 - C. Elimination of urea requires the least amount of water.
 - D. Urine formation involves filtration, reabsorption and secretion.
- Choose the most appropriate answer from the options given below:
- (1) B, C and D only
 - (2) B and D only
 - (3) A, B and C only
 - (4) A, B, C and D
- 172.** Which of the following can **not** be considered as point of difference between cortical and juxtamedullary nephrons?
- (1) their number.
 - (2) length of loop of henle.
 - (3) presence of vasa recta.
 - (4) presence of peritubular capillaries.
- 173.** Angiotensin-II is responsible for all of the following **except**;
- (1) vasoconstriction.
 - (2) increase in blood pressure.
 - (3) release of aldosterone.
 - (4) decrease reabsorption of Na^+ from renal tubule.
- 174.** Which of the following is **not** considered as component of tubular part of nephron?
- (1) PCT
 - (2) DCT
 - (3) Glomerulus
 - (4) Loop of Henle
- 175.** Under the influence of ADH maximum reabsorption of water occur in which of the following segments of nephron?
- (1) distal convoluted tubule.
 - (2) collecting duct.
 - (3) loop of henle.
 - (4) proximal convoluted tubule.
- 176.** All of the following component of blood are present in filtrate that normally enters in nephron after ultrafiltration **except**;
- (1) glucose.
 - (2) urea.
 - (3) alanine.
 - (4) albumin.
- 177.** The accumulation of urea in blood due to malfunctioning of kidney can be removed by:
- (1) kidney transplantation.
 - (2) blood donation.
 - (3) hemodialysis.
 - (4) gene therapy.
- 178.** Brush border epithelium is characteristic of:
- (1) afferent arteriole.
 - (2) collecting tube.
 - (3) proximal convoluted tubule.
 - (4) all of these
- 179.** How many nephrons are nearly present in each kidney of human beings?
- (1) One billion
 - (2) Two million
 - (3) One million
 - (4) Two billion
- 180.** Peritubular capillaries of nephron arise from;
- (1) vasa recta.
 - (2) renal artery.
 - (3) afferent arterioles.
 - (4) efferent arterioles.
- 181.** All of the following are excreted through sebum **except**;
- (1) sterols.
 - (2) hydrocarbon.
 - (3) waxes.
 - (4) NaCl.
- 182.** Glomerular filtration rate in a normal healthy man is around:
- (1) 180 mm Hg/min.
 - (2) 10 mm Hg/min.
 - (3) 60 mm Hg/min.
 - (4) 125 mm Hg/min.
- 183.** Substance like glucose, amino acid, Na^+ etc in the filtrate are reabsorbed by;
- (1) Active transport
 - (2) Passive transport
 - (3) Facilitated diffusion
 - (4) Both (1) and (2)
- 184.** Juxtaglomerular apparatus (JGA) is formed by:
- (1) PCT and efferent arteriole.
 - (2) loop of Henle and collecting duct.
 - (3) DCT and afferent arteriole.
 - (4) glomerulus and Bowman's capsule.

185. Loop of Henle is:
- (1) a hairpin loop with a descending limb followed by an ascending limb.
 - (2) a hairpin loop with an ascending limb followed by a descending limb.
 - (3) an inverted U-shaped loop with an ascending limb followed by a descending limb.
 - (4) an inverted U-shaped loop with a descending limb followed by an ascending limb.

SECTION - B

186. Dialysing fluid have the same composition as that of X except the Y;
Choose the option which fill the blanks **correctly**.

- (1) X-Plasma, Y-Nitrogenous wastes.
- (2) X-Blood, Y-Clotting factors.
- (3) X- Lymph, Y-Proteins.
- (4) X-Plasma, Y-Electrolytes

187. Given below are two statements:

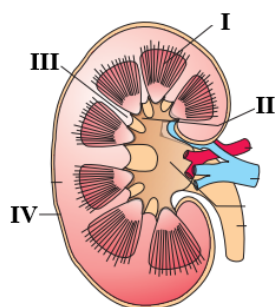
Statement I: Ammonia is the least toxic form of nitrogenous waste and requires least amount of water for its elimination.

Statement II: Protonephridia are primarily concerned with osmoregulation.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

188. Following diagram represents longitudinal section of kidney with labelled part I, II, III and IV. Which of the following is **correct** labelled parts?



- (1) I-Renal column.
- (2) II-Calyx.
- (3) III-Cortex.
- (4) IV-Renal pelvis.

189. Erythropoietin hormone is released by;

- (1) Juxtaglomerular apparatus.
- (2) Alpha cell of pancreas.
- (3) Beta cell of pancreas.
- (4) Bone marrow.

190. Which of the following hormone does leads to reabsorption of Na^+ and water from the distal part of renal tubule?

- (1) FSH
- (2) Erythropoietin
- (3) Aldosterone
- (4) Renin

191. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: During summer when body loses lot of water by evaporation, the release of ADH is suppressed.

Reason R: In the absence of ADH, water re-absorption is considerably decreased.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

192. Human urine is usually acidic because:

- (1) hydrogen ion are actively secreted into the filtrate.
- (2) the sodium transporter exchanges one hydrogen ion for each sodium ion in peritubular capillaries.
- (3) excreted plasma protein are acidic.
- (4) potassium and sodium exchange generates acidity.

193. In a healthy human being amount of urea excreted per day is:

- (1) 50-60 mg
- (2) 25-30 mg
- (3) 25-30 gm
- (4) 40-50 gm

194. Which of the following is involved in the formation of renal calculi?

- (1) oxalate.
- (2) citrate.
- (3) carbonate.
- (4) chloride ion.

195. Match List-I with List-II.

List-I		List-II	
(A)	Ultrafiltration	(I)	Henle's loop
(B)	Concentration of urine	(II)	Ureter
(C)	Transport of urine	(III)	Urinary bladder
(D)	Storage of urine	(IV)	Malpighian corpuscle

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-II, C-IV, D-I
- (4) A-IV, B-I, C-II, D-III

196. Inflammation of the glomeruli of the kidney referred to:

- (1) cystitis.
- (2) nephritis.
- (3) glomerulonephritis.
- (4) uremia.

197. On an average, ____ **X** ____ ml of blood is filtered by the kidneys per minute which constitute roughly ____ **Y** ____ of the blood pumped out by each ventricle of the heart in a minute.

Choose the option which fill the blanks **correctly**.

- | X | Y |
|---------------|-------------------|
| (1) 500–600 | 1/5 th |
| (2) 1100–1200 | 1/3 rd |
| (3) 500–600 | 1/3 rd |
| (4) 1100–1200 | 1/5 th |

198. Read the following statements (**A-E**).

- A. Terrestrial adaptation necessitated the production of lesser toxic nitrogenous wastes.
- B. Ammonia produced by metabolism is converted into urea in the liver.
- C. In some animals, urea is retained in the kidneys to regulate desired osmolarity.
- D. Nephridia remove nitrogenous wastes and regulate fluid and ionic balance.
- E. Vasa recta is absent or highly reduced in juxtamedullary nephrons.

Which of the above statements are **correct**?

- (1) A, B and E only
- (2) A, C and D only
- (3) A, B, C and D only
- (4) A, B, C, D and E

199. Which of the following sphincters cause the release of the urine?

- (1) Cardiac sphincter
- (2) Pyloric sphincter
- (3) Urethral sphincter
- (4) Oesophageal sphincter

200. Urine is stored in the urinary bladder until which of the following?

- (1) A voluntary signal from the CNS to the urinary bladder.
- (2) An involuntary signal from the CNS to the urinary bladder.
- (3) A voluntary signal from the ANS to the urinary bladder.
- (4) An involuntary signal from the ANS to the urinary bladder.

